

The Split Stage of Phase B: the split twist, its instrumentation, and the bracket-draw calibration

local-mixing project

2026-08-05

Abstract

Phase B’s job, under the 2026-08-04 reframe, is to *break the g57 structure* of a phase-A output — anti-inversion with absolute spread — rather than to mix state. We partition phase B in two parts and report the first part, now shipped: a *split stage* in which a single move — the *split twist* — splits every g57 into a (CNOT/NCNOT, 2-control AND) pair while carrying long-range *absorbed* pure-NOT twists and one cross per split. The stage runs to g57 exhaustion. On five inputs (99k–1.66M gates, 5–99% g57) it eliminates every complemented gate at a measured growth of $1+(\text{comp fraction}) (\times 1.7\text{--}2.0)$, in seconds to minutes, with every rewrite locally verified and the full circuit functionally checked. Wire canaries show twist coverage is absolute across the circuit with only the geometric end effect. A three-arm A/B over the bracket draw fixed both pathologies found on the way: the original other-half-first cascade drifted to near-full-circuit spans (midpoint crossing a constant 100%), and a naïve own-direction draw produced a 28% spike of near-zero spans. The shipped rule — twist side drawn proportionally to the circuit length remaining on that side, bracket = farthest of $k=2$ samples — lands mean span 0.57 of the circuit, midpoint crossing 71%, and a 0–5% span mass of 2.9%.

1 Context and goal

A phase-A output is a g57-dominated circuit: complemented two-control gates $a \oplus = \neg(\ell_b \wedge \ell_c)$ plus X-series conjunction gates. The g57 population is exactly the structure an inverter can lean on, so phase B opens by destroying it: every g57 must be re-spelled as ordinary conjunction material, and the re-spelling must not be locally explicable — the compensation for each local rewrite should live far away in the circuit. Phase B is therefore split in two: **part 1** (this report) breaks the structure; **part 2** is the existing phase-B menu, starting from the broken form.

2 The move

2.1 The split

One split-twist move picks a uniformly random g57 g with target w and splits it by the randomized first-failing-literal presplit (the literal shuffle is the design’s r bit). For the canonical two-control g57, $g : a \oplus = (b \vee \neg c)$ becomes either $\{a \oplus = b, a \oplus = \neg b \neg c\}$ or $\{a \oplus = \neg c, a \oplus = bc\}$: a single-control gate g_1 and a two-control AND g_2 , exact and disjointly fired. Pieces stay in place (no birth transport — g_1 must sit where the bracket forms; g_2 ’s transport is the move’s closing cross), inherit origin and litter with a fresh event, and take *opposite directions* from a fair draw. That

sibling convention is now global: every **g57** presplit anywhere in **fmix** gives its pieces alternating directions (the old independent per-piece law was a bug; pre-2026-08-05 walks do not replay).

2.2 The absorbed pure-NOT twist

With probability **p_join** the split carries a twist on wire w . Two facts make it free:

1. a gate targeting w never reads w , so it *commutes* with $X(w)$;
2. composing $X(w)$ into a 1-control gate targeting w is that gate with its control polarity flipped: $(w \oplus = \ell(b)) \cdot (w \oplus = 1) = (w \oplus = \neg \ell(b))$ — CNOT $\leftrightarrow$ NCNOT, one gate.

So for brackets $g_1 \dots h_1$ both targeting w , at any distance:

$$g_1^b \cdot S' \cdot h_1^b = g_1 \cdot X(w) \cdot S' \cdot X(w) \cdot h_1 = g_1 \cdot S \cdot h_1,$$

where b flips the bracket's control polarity and S' flips the polarity of every w -reading pin in the open segment (gates targeting w are invariant). The function is preserved with **zero synthetic gates** — strictly better than the retired free-standing NEG twist, which paid two X gates and needed a bracket-cancel tabu. This is the pure-NEG twist revived in absorbed form. The anti-inversion effect is the point: after absorption, $\{g_1^b, g_2\}$ no longer XOR to g locally — the compensation lives at h_1^b and in the flipped segment, typically across a large fraction of the circuit.

The other bracket comes from the population of gates targeting w : a **g57** (split on selection, its 1-control piece absorbs) or a CNOT/NCNOT (absorbs directly). Segment **g57s** that *read* w are force-split before their pins flip (5a). That is not needed for correctness — a polarity flip inside a complemented conjunction is exact — but it preserves the **g57+X-series** closure part 2 relies on, and it is the stage's main engine: a long twist retires every w -reading **g57** on its path.

2.3 The cross

The move closes with one ordinary cross shot from g_2 (it runs whether or not the twist landed, but not when the **p_join** coin ends the move early). Transport of the AND pieces is thus the cross machinery's job, under its normal counters and undo journal.

2.4 Layer embedding and exits

At layer 1 the split twist is a third dispatch in the twist slot (**p_split_twist**). At layer 2, **--split** arms the stage: the split twist is the *only* move running (no brake, overlay, shuffle, DB, or thermostat) until either **exit A** — no **g57** remains anywhere (in practice always this one) — or **exit B** — **--split-fail-limit** consecutive bracket failures. The boundary pins the live split-twist dispatch to zero and releases the round to the command line's phase-B parameters; **--split-stop** ends the run there instead (trial mode). The stage phase persists in the state file as a tri-state (never armed / live / ended), so resumes continue, never silently restart, and warn when a live stage is resumed without **--split**. State files are v2; v1 files load, old binaries refuse v2.

3 Instrumentation

Counters. `prims` (the picked `g57`), `hspl` (bracket splits), `segs` (segment splits), `joins`, `fails`, `xmid` (brackets straddling the midpoint), `span sum` + a 5%-bucket span histogram; a per-move `split` line carries all of them plus the move’s span.

Wire canaries (`--split-canaries`, default 256). A canary sits on a wire immediately right of an anchor gate, planted uniformly at stage start, remembering its *original* position. A twist whose bracket span covers it on its wire complements the value carried there — one flip. Canaries ride the material: on anchor death they re-anchor to the live left neighbor (the eviction must run *after* the prior partner’s unlink when both partners of a splice die — the one real bug the adversarial review caught, along with resume tri-state semantics and exit gating, all fixed and re-verified). The flips-by-original-position profile is the stage’s spread/reach deliverable.

Rank stamps. Positions and sides come from an $O(n)$ rank restamp every 8192 moves *and* on $> 25\%$ growth. Ranks are heuristics (selection, `xmid`, canary positions); correctness never depends on them — the segment itself is found by an alternating bidirectional walk.

4 Trials

Five inputs, all runs `--split --split-stop`, local verification on every rewrite, global functional checks at the boundary; every run ended by exit A with the failure limit untouched.

input	gates	g57 frac	moves	end size	growth	time
gadgetized n128	99,016	5.2%	1,462	106,000	$\times 1.07$	1.5s
cg1 slow-anneal	1,020,340	69.2%	5,495	1,731,515	$\times 1.70$	129s
cg1 fast-anneal	1,016,535	16.1%	3,788	1,184,051	$\times 1.16$	74s
cg1_phaseA5 ($p_{\text{join}}=1$)	64,782	73.5%	2,713	115,629	$\times 1.78$	4s
g57A phaseA ($p_{\text{join}}=1$)	1,664,636	83.5%	4,818	3,060,082	$\times 1.84$	293s
nR20_mixed ($p_{\text{join}}=1$)	179,132	99.3%	5,357	362,995	$\times 2.03$	27s

Three regularities, all predicted by the design: **(i) growth $\approx 1 + \text{comp fraction}$** — the pair split *is* the growth; cross ladders are net noise. **(ii) segment splits dominate:** $\sim 99\%$ of `g57` depletion arrives through 5a — the stage is a few thousand large sweeps, not a per-gate crawl. **(iii) brackets never starve:** every split mints a permanent 1-control bracket, so failures stay near zero and exit is always by exhaustion.

Gate makeup. The cleanest case, `nR20_mixed` (99.3% true `g57`, no 1-control gates at all on input) exits as: CNOT 24.6% / NCNOT 24.6% (dead-even — the polarity churn of the absorbed NOTs), 2-control ANDs at a 1:2:1 negation-count ratio (the binomial signature of the presplit composed with uniform polarity churn), a $\sim 2\%$ cross-ladder tail, zero comp. X gates pass through untouched in absolute count on every input.

5 Coverage: the canaries

Split-stage NOT-twist coverage: canary flips by original wire-segment position
(line = decile mean, band = ± 1 std, dots = individual canaries)

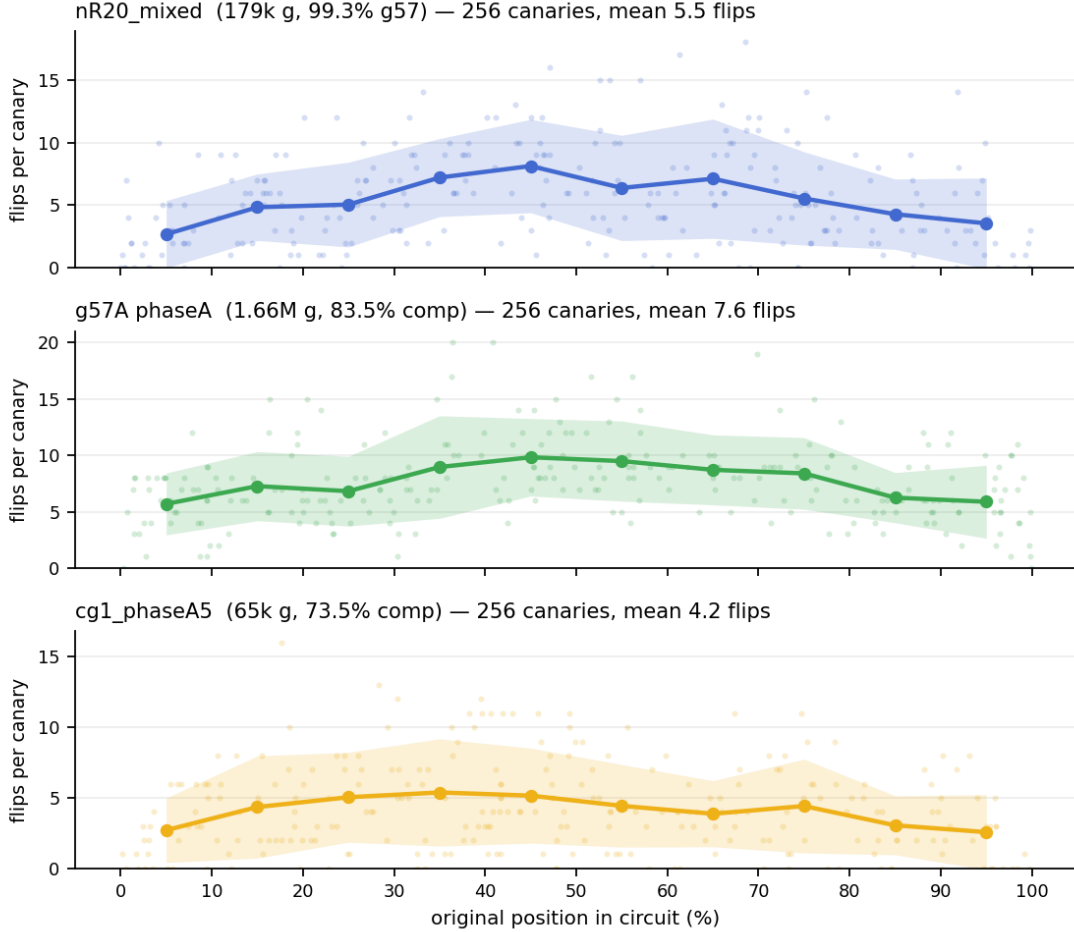


Figure 1: Flips per canary by *original* position: decile mean (line), ± 1 std (band), individual canaries (dots); the three $p_{\text{join}}=1$ phase-A runs. A broad mid-circuit plateau at roughly $2\times$ the edge coverage — the $2x(1-x)$ geometry of a span between two near-uniform endpoints — with no asymmetry and no origin clustering. Std $\approx$ mean: spans are few and long, so per-canary counts are overdispersed.

Reach is absolute up to the unavoidable end effect. If flatter *edge* coverage is ever wanted, the lever is biasing bracket choice toward the ends, not more twists.

6 Calibrating the bracket draw

The original draw was a cascade: prefer a g57 in the *other half* of the circuit, then a CNOT there, then the same-half classes. It worked — but `xmid` sat at a constant 100%: an other-half bracket

essentially always exists, so the midpoint-crossing counter was a target, not a measurement, and an overshoot. Two redesign steps, measured on `nR20_mixed` at equal seed (Figure 2):

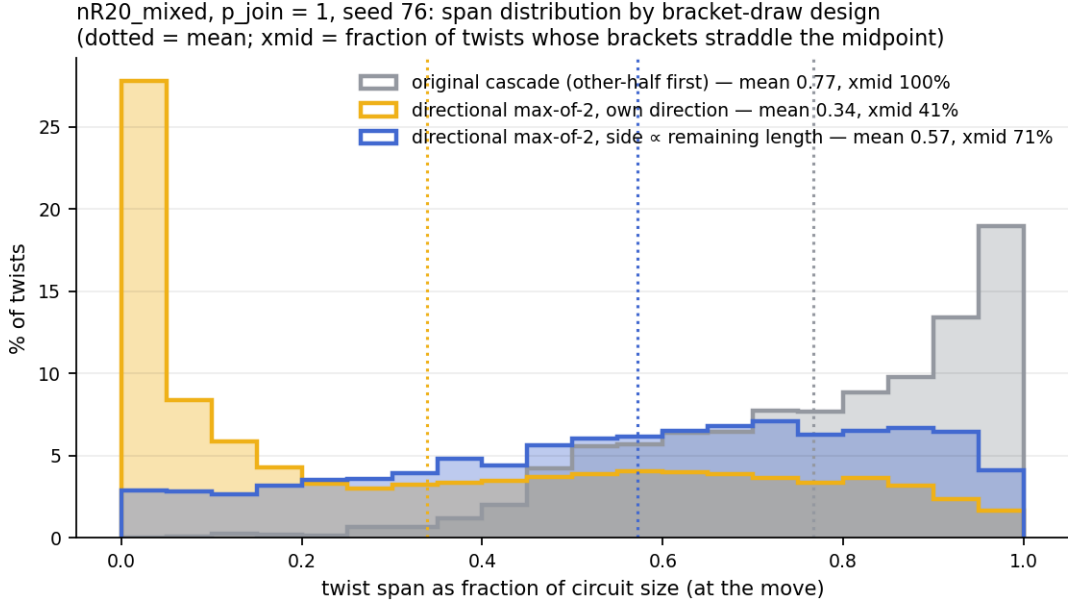


Figure 2: Twist span as a fraction of the circuit, three bracket-draw designs on `nR20_mixed` ($p_{\text{join}}=1$, same seed).

arm	mean span	0–5% mass	xmid	fails	moves
other-half cascade ($k=0$)	0.77	0.03%	100%	0	3,811
own-direction, max-of-2	0.34	27.8%	41%	137	6,417
side $\propto$ length, max-of-2 (shipped)	0.57	2.9%	71%	1	5,357

The cascade drifts long. Its spans are far longer than uniform other-half placement predicts (mode 95–100%). The histogram exposed a feedback loop: segment sweeps clear the middle first, surviving `g57s` cluster at the edges, and edge-picked primaries paired with other-half brackets span nearly the whole circuit — self-reinforcing as the stage proceeds.

Own-direction spikes short. Restricting candidates to the picked `g57`’s own stored direction and taking the farthest of $k=2$ uniform samples lands the design mean ($\frac{2}{3} \cdot \mathbb{E}[\text{available run}] \approx 0.34$) — but 28% of twists fall in the 0–5% bucket: primaries picked near the edge their direction points at have almost no run to use.

Shipped: side drawn $\propto$ remaining length. The twist side is drawn with probability proportional to the circuit length remaining on it, so a side is picked exactly as rarely as it is short — a tiny span now needs a short side *and* the coin to pick it (squared suppression). The 0–5% mass collapses to 2.9%, the mean lands at 0.57 (above the naïve $4/9$ because edge-clustered survivors are sent sweeping across the circuit), midpoint crossing settles at an honest 71%, failures essentially vanish, and the canary deciles show the best edge coverage of the three arms (3.4... 9.4... 3.4). The knob `--split-reach-k` remains the length-bias dial (1 = uniform on the drawn side, $2 \approx \frac{2}{3}$ of the run, $3 \approx \frac{3}{4}$; 0 = the legacy cascade, kept as an A/B arm).

7 Shipped parameters and launch recipe

knob	default	meaning
<code>--split</code>	off	arm the stage (forces the split twist as the only move)
<code>--split-stop</code>	off	end the run at the stage boundary (trial mode)
<code>--p-join</code>	0.8	probability a split carries the twist + cross (trials ran 1.0)
<code>--split-reach-k</code>	2	length bias of the bracket draw (0 = legacy cascade)
<code>--split-fail-limit</code>	100	exit B threshold (never approached in practice)
<code>--split-canaries</code>	256	coverage monitors, reported at the boundary

Launch hygiene unchanged: export `FMIX_STOP_FLAG/FMIX_DUMP_FLAG`, budget `--moves` generously (the stage self-terminates; one move per split twist, so a comfortable multiple of the `g57` count), pin every measured knob, and keep `--p-db 0 --p-comp 0 --p-any 0` for a stage-only invocation so no store is opened. `--split` and `--profile` are mutually exclusive; run part 2 as its own invocation.

8 Open items

Part-2 calibration starts from the split form ($\times 1.7$ – 2.0 the phase-A size, zero comp, 50/50 CNOT/NCNOT); the thermostat target and move budget should be sized against that, and old descent-free calibrations do not transfer. The wide-conjunction tail of some phase-A recipes (up to `w32`) passes through the stage untouched except for pin flips; per-gate exhaustive verification is capped at width < 16 (wide flips remain covered by the global functional check). Artifacts: spec docs/FMIX_SPLIT_TWIST.md, measurements and scripts reports/split_trials.20260805/.